

LEFSCHETZ FIBRATIONS

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ABSTRACT. IAS/PCMI lecture notes.

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¹Based on lectures by Hisaaki Endo [6].

Part 1. Lefschetz Fibrations²

1. LEFSCHETZ FIBRATIONS AND THEIR MONODROMY

A Lefschetz fibration is a surface bundle away from finitely many complex-Morse singularities, and its monodromy is recorded by right-handed Dehn twists.

1.1. Fibrations, Monodromy, and Vanishing Cycles. Throughout, X and Σ are connected compact manifolds of dimensions 4 and 2, $f: X \rightarrow \Sigma$ is surjective, $\mathcal{C} := f(\text{Crit}(f))$, and

$$\begin{aligned} \Sigma_g &:= \text{the closed oriented surface of genus } g, & \mathcal{M}_g &:= \pi_0(\text{Diff}^+(\Sigma_g)), \\ t_c &:= \text{the right-handed Dehn twist about } c. \end{aligned}$$

The group $\mathcal{M}_g$ acts on the right, diffeomorphisms are identified with their mapping classes, and every curve used as a vanishing cycle is homotopically essential [6].

Definition 1.1.1 (Lefschetz fibration, [6, 16]). The map $f: X \rightarrow \Sigma$ is a *Lefschetz fibration* when

$$f^{-1}(\partial\Sigma) = \partial X, \quad \text{Crit}(f) \subseteq \text{Int}(X), \quad \text{no fiber contains an embedded } (-1)\text{-sphere},$$

and every $p \in \text{Crit}(f)$ admits orientation-preserving complex charts in which

$$(1) \quad (\psi \circ f \circ \varphi^{-1})(z, w) = zw.$$

The manifolds X, Σ , the map f , and $f^{-1}(q)$ are respectively the *total space*, *base*, *projection*, and the fiber over q ; the fiber is singular exactly when $q \in \mathcal{C}$.

Proposition 1.1.2 (Regular-fiber bundle, [4]). *For every Lefschetz fibration,*

$$\#\text{Crit}(f) < \infty, \quad f|_{f^{-1}(\Sigma - \mathcal{C})}: f^{-1}(\Sigma - \mathcal{C}) \rightarrow \Sigma - \mathcal{C} \text{ is a smooth fiber bundle.}$$

The fibration has genus g when its regular fiber is Σ_g .

Proof. The local model isolates the critical points, and Ehresmann's theorem controls the complement.

$$(1) \implies \text{Crit}(f) \subseteq X \text{ is discrete}, \quad X \text{ compact} \implies \#\text{Crit}(f) < \infty.$$

$$f|_{f^{-1}(\Sigma - \mathcal{C})} \text{ is a proper submersion} \implies f^{-1}(\Sigma - \mathcal{C}) \rightarrow \Sigma - \mathcal{C} \text{ is a smooth fiber bundle.}$$

Connectedness of $\Sigma - \mathcal{C}$ makes every regular fiber diffeomorphic to Σ_g . □

Theorem 1.1.3 (Critical-value separation, [6]). *Every Lefschetz fibration can be deformed through smooth maps, preserving the local models (1), so that*

$$f|_{\text{Crit}(f)}: \text{Crit}(f) \rightarrow \Sigma \text{ is injective.}$$

Proof. Separate one coincident pair in an interior chart and iterate over the finite critical set. For $p \neq p'$ with $f(p) = f(p')$, choose χ supported in a source chart with compact closure in $\text{Int}(X)$, equal to 1 near p , and set

$$(\psi \circ f_s \circ \varphi^{-1})(z, w) := zw + s\varepsilon\chi(|z|^2 + |w|^2), \quad 0 \leq s \leq 1.$$

$$\left(\text{chart with one critical point} \right) \xrightarrow{\varepsilon\chi} \left(\text{chart with two critical points} \right), \quad f_\varepsilon(p) = f(p) + \varepsilon \neq f_\varepsilon(p').$$

$$\text{supp}(d\chi) \cap \text{Crit}(f) = \emptyset, \quad \text{supp}(d\chi) \text{ compact} \implies |\varepsilon| \text{ small} \implies df_s \text{ surjective there.}$$

$$\chi = 1 \text{ near } p \implies f_s = zw + s\varepsilon, \quad \chi = 0 \text{ off the chart} \implies f_s = f.$$

Choosing ε away from the remaining critical values and repeating finitely often separates all critical values without changing any Lefschetz germ. □

Every Lefschetz fibration below satisfies Theorem 1.1.3.

²Based on lectures by Hisaaki Endo [6].

Definition 1.1.4 (Achiral Lefschetz fibration, [6]). If the source chart need not preserve orientation, f is an *achiral Lefschetz fibration*; its local models are

$$(z_1, z_2) \mapsto z_1 z_2, \quad (z_1, z_2) \mapsto \bar{z}_1 z_2.$$

Fix $q_0 \in \Sigma - \mathcal{C}$, write $F_{q_0} := f^{-1}(q_0)$, and choose an orientation-preserving identification $\Phi: \Sigma_g \rightarrow F_{q_0}$. For a based loop $a: [0, 1] \rightarrow \Sigma - \mathcal{C}$, write

$$a^*X := \{(t, p) \in [0, 1] \times X \mid a(t) = f(p)\}, \quad \Psi: [0, 1] \times \Sigma_g \xrightarrow{\cong} a^*X, \quad \Psi(0, x) = (0, \Phi(x)),$$

and let $\Psi_1: \Sigma_g \rightarrow f^{-1}(q_0)$ be the endpoint transport.

Definition 1.1.5 (Monodromy representation, [6, 17]). The *monodromy* along $[a]$ and the *monodromy representation* are

$$\rho_f([a]) := [\Phi^{-1} \circ \Psi_1] \in \mathcal{M}_g, \quad \rho_f: \pi_1(\Sigma - \mathcal{C}, q_0) \longrightarrow \mathcal{M}_g.$$

Theorem 1.1.6 (Well-definedness and functoriality of monodromy, [6]). *The class $\rho_f([a])$ is independent of the representative a and the trivialization Ψ , and*

$$\rho_f(\alpha\beta) = \rho_f(\alpha)\rho_f(\beta).$$

If Φ' is another identification and $\phi := [(\Phi')^{-1} \circ \Phi]$, then

$$\rho'_f([a]) = \phi^{-1} \cdot \rho_f([a]) \cdot \phi.$$

Proof. Bundle transport over a contractible parameter space makes the choices unique up to fiber isotopy.

$$\Psi, \Psi' \xrightarrow{1.1.5} (\Psi^{-1}\Psi')_0 = \text{id} \implies [\Phi^{-1}\Psi_1] = [\Phi^{-1}\Psi'_1],$$

$$H: a \simeq a' \xrightarrow{1.1.2} H^*X \longrightarrow [0, 1]^2 \text{ trivial} \implies \rho_f([a]) = \rho_f([a']).$$

Stacking transports and changing the fiber identification are represented by

$$\left(F_{q_0} \xrightarrow{P_\alpha} F_{q_0} \xrightarrow{P_\beta} F_{q_0} \right) \xrightarrow{\text{stacking}} P_{\alpha\beta} = P_\beta \circ P_\alpha, \quad \rho_f(\alpha\beta) = \rho_f(\alpha)\rho_f(\beta).$$

$$\left(\begin{array}{ccc} \Sigma_g & \xrightarrow{\phi} & \Sigma_g \\ & \searrow \Phi & \nearrow (\Phi')^{-1} \\ & F_{q_0} & \end{array} \right), \quad \phi = (\Phi')^{-1} \circ \Phi \xrightarrow{\text{right action}} \rho'_f([a]) = \phi^{-1} \rho_f([a]) \phi.$$

These identities establish independence, multiplicativity, and the conjugation rule. $\square$

Definition 1.1.7 (Complex Morse local model, [6]). For $Q(z, w) := zw$, projection to the first coordinate gives

$$Q^{-1}(\varepsilon) = \{w = \varepsilon z^{-1}\} \xrightarrow{\text{pr}_1} \mathbb{C} - \{0\}, \quad Q^{-1}(0) = \{w = 0\} \cup \{z = 0\}.$$

With $T := \{|z| = |w|\}$, the *vanishing thimble* and *vanishing cycle* are

$$\Delta_\varepsilon := \bigcup_{0 \leq s \leq 1} (Q^{-1}(s\varepsilon) \cap T), \quad \partial\Delta_\varepsilon = Q^{-1}(\varepsilon) \cap T \xrightarrow{\text{pr}_1} \{|z| = \sqrt{|\varepsilon|}\}.$$

Set

$$K := \{|z| \leq 2, |w| \leq 2\}, \quad A_i := \{(\tau, i) \mid |\tau| \geq 2\}, \quad A := A_1 \cup A_2.$$

Proposition 1.1.8 (Exterior triviality of the nodal model, [6]). *For $D^2 := \{|u| \leq 1\}$, the restriction*

$$Q: Q^{-1}(D^2) - \text{Int}(K) \longrightarrow D^2$$

is a trivial fiber bundle with fiber A .

Proof. The two components of A give global coordinates on the two ends of every fiber.

$$F(u, \tau, i) := \begin{cases} (\tau, u/\tau), & i = 1, \\ (u/\tau, \tau), & i = 2, \end{cases} \quad F^{-1}(z, w) := \begin{cases} (zw, z, 1), & |z| \geq 2, \\ (zw, w, 2), & |w| \geq 2. \end{cases}$$

$$|zw| \leq 1, \quad (z, w) \notin \text{Int}(K) \implies (|z| \geq 2) \dot{\vee} (|w| \geq 2),$$

$$F^{-1} \circ F = \text{id}_{D^2 \times A}, \quad F \circ F^{-1} = \text{id}_{Q^{-1}(D^2) - \text{Int}(K)}, \quad Q \circ F = \text{pr}_1 \implies F: D^2 \times A \xrightarrow{\cong} Q^{-1}(D^2) - \text{Int}(K).$$

Thus F is the required fiber-preserving trivialization. $\square$

Although $Q^{-1}(D^2)$ is noncompact, Q satisfies the local and boundary conditions of Definition 1.1.1 and will be used only as a local model. Set

$$X := Q^{-1}(D^2), \quad f := Q|_X, \quad q_0 := 1, \quad a(t) := e^{2\pi\sqrt{-1}t}.$$

Proposition 1.1.9 (Local monodromy trivialization, [6]). *Fix $0 < \varepsilon < \frac{1}{2}$, and let $\rho: \mathbb{R} \rightarrow [0, 2\pi]$ be smooth and nondecreasing, with $\rho = 0$ below $1 - \varepsilon$ and $\rho = 2\pi$ above $1 + \varepsilon$. The map*

$$\Psi(t, x) := \begin{cases} (t, F(a(t), \text{pr}_2(F^{-1}(x)))) & x \in f^{-1}(q_0) - \text{Int}(K), \\ (t, (ze^{(2\pi - \rho(|z|))\sqrt{-1}t}, we^{\rho(|z|)\sqrt{-1}t})) & x = (z, w) \in f^{-1}(q_0) \cap K, \end{cases}$$

*is a trivialization of a^*f .*

Proof. The two formulas lie in the required fibers and agree along the gluing region.

$$Q\left(ze^{(2\pi - \rho(|z|))\sqrt{-1}t}, we^{\rho(|z|)\sqrt{-1}t}\right) = e^{2\pi\sqrt{-1}t} = a(t), \quad Q \circ F(a(t), \cdot) = a(t).$$

$$|z| \geq 2 \implies \rho(|z|) = 2\pi \implies (z, a(t)w) = F(a(t), (z, 1)),$$

$$|w| \geq 2 \implies |z| \leq \frac{1}{2} \implies \rho(|z|) = 0 \implies (a(t)z, w) = F(a(t), (w, 2)).$$

$$z = re^{\sqrt{-1}\theta} \mapsto re^{\sqrt{-1}(\theta + (2\pi - \rho(r))t)}, \quad \Psi(0, x) = (0, x).$$

Each time slice is a diffeomorphism, so the glued map is the asserted trivialization. $\square$

Theorem 1.1.10 (Picard–Lefschetz local monodromy, [6]). *For $\Psi_1(x) := \text{pr}_2(\Psi(1, x))$ and $f^{-1}(q_0) = \{(z, z^{-1}) \mid z \neq 0\}$,*

$$\rho_f([a]) = [\Psi_1], \quad (\text{pr}_1 \circ \Psi_1 \circ \text{pr}_1^{-1})(z) = \begin{cases} z, & |z| \geq 2 \text{ or } |z| \leq \frac{1}{2}, \\ ze^{-\rho(|z|)\sqrt{-1}}, & \frac{1}{2} \leq |z| \leq 2. \end{cases}$$

By Figure ??, this is the compactly supported right-handed Dehn twist about the vanishing cycle $Q^{-1}(1) \cap T$.

Proof. The endpoint of the explicit trivialization is the identity off the local core and a damped rotation on it.

$$\Psi(0, x) = (0, x) \xrightarrow{1.1.5} \rho_f([a]) = [\Psi_1], \quad a(1) = a(0) \xrightarrow{1.1.9} \Psi_1 = \text{id off } K,$$

$$x = (z, z^{-1}) \in K \xrightarrow{1.1.9} \text{pr}_1 \Psi_1(x) = ze^{(2\pi - \rho(|z|))\sqrt{-1}} = ze^{-\rho(|z|)\sqrt{-1}}.$$

By Figure ??, the damped rotation winds a transverse arc once and gives the right-handed twist. $\square$

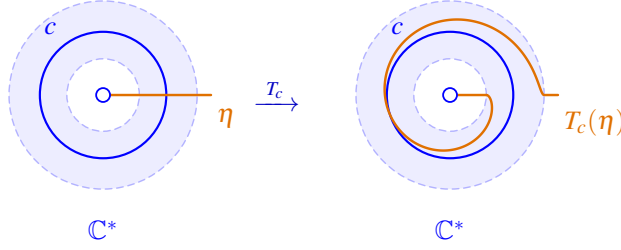


FIGURE 1. The right-handed Dehn twist T_c about $c = \{|z| = 1\} \subset \mathbb{C}^*$ sends the transverse arc η to an arc winding once around c .

Definition 1.1.11 (Hurwitz system, [6]). Let $\mathcal{C} = \{q_1, \dots, q_n\} \subseteq \text{Int}(D^2)$ and $q_0 \in D^2 - \mathcal{C}$. A *Hurwitz system* is a counterclockwise ordered tuple $(\gamma_1, \dots, \gamma_n)$ of simple paths from q_0 to the q_i , intersecting only at q_0 . If m_i is the positive meridian of q_i and δ_i is the part of γ_i ending at m_i , then

$$a_i := \delta_i \cdot m_i \cdot \delta_i^{-1}, \quad \pi_1(D^2 - \mathcal{C}, q_0) \cong F_n = \langle [a_1], \dots, [a_n] \rangle.$$

Definition 1.1.12 (Monodromy factorization, [6, 16]). For a genus- g Lefschetz fibration over D^2 , Theorem 1.1.10 assigns vanishing cycles $c_i \subseteq \Sigma_g$ satisfying

$$\rho_f([a_i]) = t_{c_i}.$$

The tuple and its product

$$(t_{c_1}, \dots, t_{c_n}), \quad t_{c_1} \cdots t_{c_n},$$

are respectively the *monodromy factorization* and *total monodromy* of f .

1.2. Constructions and Classification. Positive twist factorizations are realized over D^2 ; over S^2 , the total monodromy is the identity, and the resulting representation classifies the fibration under the standard hypotheses.

Theorem 1.2.1 (Disk realization, [6]). *Let $c_1, \dots, c_n \subseteq \Sigma_g$ be homotopically essential simple closed curves. There is a genus- g Lefschetz fibration $f: X \rightarrow D^2$, a Hurwitz system, and an orientation-preserving identification $\Phi: \Sigma_g \rightarrow f^{-1}(q_0)$ whose monodromy factorization is*

$$(t_{c_1}, \dots, t_{c_n}).$$

Proof. Graft one compact complex-Morse core into a product bundle along collars of each prescribed curve. For $R > 2$, put

$$K_R := \{|z| \leq R, |w| \leq R\}, \quad W_R := Q^{-1}(D^2) \cap K_R, \quad A_R := \{(\tau, i) \mid 2 \leq |\tau| \leq R, i = 1, 2\}.$$

For one curve c , let $\widehat{\Sigma} := \Sigma_g - \text{Int}(v(c))$ and identify A_R with collars of $\partial\widehat{\Sigma}$. After rounding corners, form

$$\left(\begin{array}{ccc} D^2 \times A_R & \xrightarrow{F} & W_R \\ \downarrow & & \downarrow \\ D^2 \times \widehat{\Sigma} & \hookrightarrow & X \end{array} \right) \xrightarrow{\text{fiberwise pushout}} f^{-1}(z) = \widehat{\Sigma} \cup_{A_R} (Q^{-1}(z) \cap K_R) \cong \Sigma_g.$$

$$Q \circ F = \text{pr}_1, \quad \text{Crit}(f) = \{(0, 0)\}, \quad \rho_f([a_1]) = t_c \xrightarrow{1.1.10} c \text{ is the vanishing cycle.}$$

For general n , cut along the Hurwitz paths and graft the cores over pairwise disjoint disks D_i .

$$\left(\begin{array}{c} \text{Diagram of a genus-}g \text{ surface with } n \text{ disks } D_i \text{ and paths } \gamma_i \end{array} \right), \quad B := D^2 - \bigcup_{i=1}^n (\text{Int}(D_i) \cup \gamma_i).$$

$$f^{-1}(B) \cong B \times \Sigma_g, \quad \text{Crit}(f) = \{p_1, \dots, p_n\}, \quad \rho_f([a_i]) = t_{c_i} \quad (1 \leq i \leq n).$$

Essentiality gives relative minimality, so the glued map is the required Lefschetz fibration. $\square$

For a Hurwitz system on S^2 , the positive meridians satisfy

$$(2) \quad [a_1] \cdots [a_n] = 1 \xrightarrow{1.1.5} t_{c_1} \cdots t_{c_n} = 1 \quad \text{in } \mathcal{M}_g.$$

Theorem 1.2.2 (Sphere realization, [6]). *Homotopically essential curves $c_1, \dots, c_n \subseteq \Sigma_g$ satisfying (2) occur as the monodromy factorization of a genus- g Lefschetz fibration $f: X \rightarrow S^2$.*

Proof. Realize the factorization over a disk and cap its trivial boundary bundle. By Theorem 1.2.1, obtain $f_0: X_0 \rightarrow D^2$; if $h \in \text{Diff}^+(\Sigma_g)$ represents its boundary monodromy, then

$$[h] = t_{c_1} \cdots t_{c_n} = 1 \in \mathcal{M}_g \implies h \simeq \text{id}_{\Sigma_g} \implies f_0^{-1}(\partial D^2) \cong S^1 \times \Sigma_g.$$

$$\left(\begin{array}{ccc} S^1 \times \Sigma_g & \hookrightarrow & D^2 \times \Sigma_g \\ \downarrow & & \downarrow \\ X_0 & \hookrightarrow & X \end{array} \right) \rightarrow \left(\begin{array}{ccc} S^1 & \hookrightarrow & D^2 \\ \downarrow & & \downarrow \\ D^2 & \hookrightarrow & S^2 \end{array} \right), \quad t_{c_1} \cdots t_{c_n} = 1.$$

$$\text{Crit}(f) = \text{Crit}(f_0), \quad \rho_f([a_i]) = \rho_{f_0}([a_i]) = t_{c_i}, \quad f|_{D^2 \times \Sigma_g} = \text{pr}_1.$$

The nonsingular cap preserves the ordered vanishing cycles and yields the required fibration over S^2 . $\square$

Definition 1.2.3 (Isomorphism and monodromy equivalence, [16, 17]). Genus- g Lefschetz fibrations $f_i: X_i \rightarrow \Sigma$ ($i = 0, 1$) are *isomorphic* when orientation-preserving diffeomorphisms Θ, θ satisfy

$$\begin{array}{ccc} X_0 & \xrightarrow{\Theta} & X_1 \\ f_0 \downarrow & & \downarrow f_1 \\ \Sigma & \xrightarrow{\theta} & \Sigma \end{array}$$

and homomorphisms $\rho_i: \pi_1(\Sigma - \mathcal{C}_i, q_i^0) \rightarrow \mathcal{M}_g$ are *equivalent* when an orientation-preserving $\theta: (\Sigma, \mathcal{C}_0, q_0^0) \rightarrow (\Sigma, \mathcal{C}_1, q_1^0)$ and $\phi \in \mathcal{M}_g$ satisfy

$$(\rho_1 \circ \theta_*)(\alpha) = \phi^{-1} \cdot \rho_0(\alpha) \cdot \phi.$$

Theorem 1.2.4 (Kas–Matsumoto classification, [16, 17]). *Let f_0, f_1 be genus- g Lefschetz fibrations over Σ , and assume $g \geq 2$, or $\partial \Sigma \neq \emptyset$, or $\mathcal{C}_0 \neq \emptyset$. Then*

$$f_0 \cong f_1 \iff \rho_0 \text{ and } \rho_1 \text{ are equivalent.}$$

Proof. An isomorphism restricts to the regular bundles, while equivalent monodromy reconstructs the same nodal attachments.

$$f_0 \cong f_1 \implies (\rho_1 \circ \theta_*)(\alpha) = \phi^{-1} \cdot \rho_0(\alpha) \cdot \phi.$$

$$\rho_1 \circ \theta_* = \phi^{-1} \cdot \rho_0 \cdot \phi \xrightarrow{1.1.2} \left(\begin{array}{ccc} X_0^\circ & \xrightarrow{\Theta^\circ} & X_1^\circ \\ f_0 \downarrow & & \downarrow f_1 \\ \Sigma - \mathcal{C}_0 & \xrightarrow{\theta} & \Sigma - \mathcal{C}_1 \end{array} \right).$$

$$\rho_0([a_i]) = t_{c_i}, \quad \rho_1([\theta(a_i)]) = \phi^{-1} \cdot t_{c_i} \cdot \phi \xrightarrow{1.1.10} \Theta^\circ \text{ extends across the } i\text{-th nodal core.}$$

The Kas–Matsumoto extension theorem and the stated hypotheses extend Θ° across every singular fiber, proving $f_0 \cong f_1$. $\square$

Definition 1.2.5 (Elementary Hurwitz transformation, [6]). For a Hurwitz system on D^2 or S^2 , an *elementary transformation* replaces adjacent paths (γ_i, γ_{i+1}) by the positive half-twisted pair $(\tilde{\gamma}_i, \tilde{\gamma}_{i+1})$ of Figure ??; its inverse is also elementary.

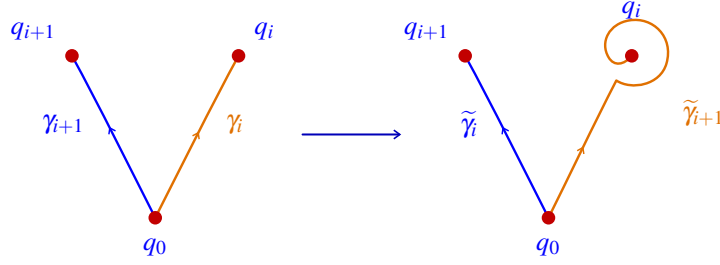


FIGURE 2. An elementary transformation of two adjacent paths in a Hurwitz system.

Theorem 1.2.6 (Connectivity of Hurwitz systems, [6]). *Any two Hurwitz systems for $(\Sigma, \mathcal{C}, q_0)$, with $\Sigma = D^2$ or S^2 , are related by finitely many elementary transformations, their inverses, and cyclic permutations.*

Proof. Work up to ambient isotopy relative to $\mathcal{C} \cup \{q_0\}$, fix $\delta := \gamma'_1$, and put δ transverse to $|\Gamma| := \bigcup_i \gamma_i$. Set

$$i_\delta(\Gamma) := \#((\delta \cap |\Gamma|) - (\mathcal{C} \cup \{q_0\})).$$

An outermost component of $\delta - |\Gamma|$ cuts off a disk whose Γ -branches are consecutive. Slide them across one at a time by

$$\left(\begin{array}{c} \text{Diagram of paths } \gamma_i, \gamma_{i+1} \text{ meeting at } q_0 \end{array} \right) \xrightarrow{\text{elem. } \pm 1} \left(\begin{array}{c} \text{Diagram of paths } \tilde{\gamma}_i, \tilde{\gamma}_{i+1} \text{ meeting at } q_0 \end{array} \right) \implies i_\delta(\tilde{\Gamma}) < i_\delta(\Gamma).$$

At $i_\delta = 0$, let $\gamma \in \Gamma$ end at $\delta(1)$. Adjacent slides empty the disk bounded by $\gamma \cup \delta$, so $\gamma \simeq \delta$ relative to its endpoints. Cut along this common path, induct on $\#\mathcal{C}$, and finish by a cyclic shift at q_0 . $\square$

Definition 1.2.7 (Hurwitz equivalence, [6]). The *Hurwitz moves*, *simultaneous conjugations*, and *cyclic moves* are

$$\begin{aligned} (\dots, t_{c_i}, t_{c_{i+1}}, \dots) &\longleftrightarrow (\dots, t_{c_i} t_{c_{i+1}} t_{c_i}^{-1}, t_{c_i}, \dots), \\ (t_{c_1}, \dots, t_{c_n}) &\longleftrightarrow (\phi^{-1} t_{c_1} \phi, \dots, \phi^{-1} t_{c_n} \phi), \quad \phi \in \mathcal{M}_g, \\ (t_{c_1}, \dots, t_{c_{n-1}}, t_{c_n}) &\longleftrightarrow (t_{c_n}, t_{c_1}, \dots, t_{c_{n-1}}). \end{aligned}$$

Two tuples are *Hurwitz equivalent*, written $\sim_H$, when finitely many such moves connect them; the conjugation order follows from the right action fixed in Subsection 1.1.

Corollary 1.2.8 (Monodromy classification over the sphere, [6]). *For $g \geq 2$ or $n \geq 1$, there is a bijection $\{f: X \rightarrow S^2 \mid f \text{ has genus } g \text{ and } n \text{ singular fibers}\} / \cong \longleftrightarrow \{(t_{c_1}, \dots, t_{c_n}) \mid c_i \text{ essential}, t_{c_1} \cdots t_{c_n} = 1\} / \sim_H$.*

Proof. The classification theorem identifies isomorphism with equivalence of representations, while Figure ?? accounts for changes of Hurwitz system.

$$f_0 \cong f_1 \xLeftrightarrow{1.2.4} \rho_0 \text{ is equivalent to } \rho_1 \xLeftrightarrow{1.2.6, 1.2.7} (t_{c_1}, \dots, t_{c_n}) \sim_H (t_{d_1}, \dots, t_{d_n}).$$

Every identity factorization is realized by Theorem 1.2.2, so the correspondence is bijective. $\square$

Let $c_1, \dots, c_{2g+1} \subseteq \Sigma_g$ be the chain of Figure 3, so

$$i(c_j, c_k) = \begin{cases} 1, & |j - k| = 1, \\ 0, & |j - k| \geq 2. \end{cases}$$

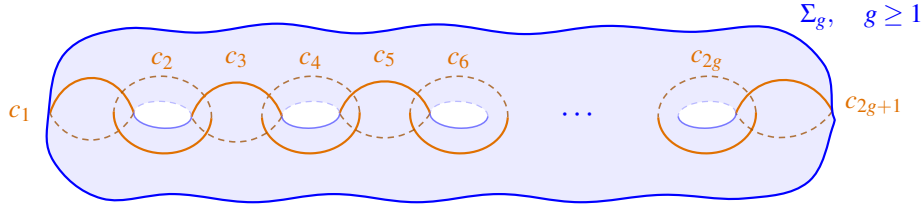


FIGURE 3. A standard chain of simple closed curves $c_1, \dots, c_{2g+1}$ on the genus- g surface Σ_g .

Definition 1.2.9 (Standard monodromy sequences, [6, 17]). Write $t_i := t_{c_i}$. Concatenation and repetition of tuples define

$$\begin{aligned} \varphi_0 &:= (t_1, t_2, \dots, t_{2g+1}, t_{2g+1}, \dots, t_2, t_1)^2, & \varphi_1 &:= (t_1, t_2, \dots, t_{2g+1})^{2g+2}, \\ \varphi_2 &:= (t_1, t_2, \dots, t_{2g})^{4g+2}. \end{aligned}$$

The chain relations give

$$\prod \varphi_0 = \prod \varphi_1 = \prod \varphi_2 = 1 \xrightarrow{1,2,2} f_i: X_i \longrightarrow S^2 \quad (i = 0, 1, 2),$$

with monodromy factorizations φ_i .

Theorem 1.2.10 (Hurwitz equivalences of the standard factorizations, [6]). *The standard words satisfy*

$$g = 1 \implies \varphi_0 \sim_H \varphi_1 \sim_H \varphi_2, \quad \varphi_0^{g+1} \sim_H \varphi_1^2, \quad \varphi_1^2 \sim_H \varphi_2 \varphi_0.$$

Proof. Set

$$\begin{aligned} m &:= 2g + 2, & A &:= (t_1, \dots, t_{m-1}), & B &:= (t_{m-1}, \dots, t_1), \\ C &:= (t_1, \dots, t_{m-2}), & T &:= AB, & P &:= C^{m-1}, & Q &:= BA. \end{aligned}$$

Thus

$$(\varphi_0, \varphi_1, \varphi_2) = (T^2, A^m, C^{2(m-1)}).$$

The local H1- and H2-moves are the commuting and Artin braid relations, while H3 slides T through every chain twist.

$$\left(\begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} \right) \stackrel{H2}{\sim} \left(\begin{array}{c} \diagdown \diagup \\ \diagup \diagdown \end{array} \right).$$

For $g = 1$, put $x := t_1 = t_3$ and $y := t_2$; the same local move gives the entire equivalence [19].

$$\left(\begin{array}{c} \diagup \diagdown \diagup \diagdown \\ \diagdown \diagup \diagdown \diagup \end{array} \right)^2 \stackrel{g=1}{\sim} \left(\begin{array}{c} \diagup \diagdown \diagdown \diagup \\ \diagdown \diagup \diagup \diagdown \end{array} \right)^2, \quad \varphi_0 = \varphi_1 \sim_H \varphi_2.$$

For the general movies, six visible strands display the m -strand pattern; inserting the parallel middle strands and their staircase crossings gives the same move for every m .

$$\begin{aligned} A &= \left(\begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} \right), & B &= \left(\begin{array}{c} \diagdown \diagup \\ \diagup \diagdown \end{array} \right), & C &= \left(\begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} \right), \\ T &= \left(\begin{array}{c} \diagup \diagdown \diagup \diagdown \\ \diagdown \diagup \diagdown \diagup \end{array} \right), & Q &= \left(\begin{array}{c} \diagdown \diagup \diagdown \diagup \\ \diagup \diagdown \diagup \diagdown \end{array} \right). \end{aligned}$$

Writing

$$\alpha := \sigma_1 \cdots \sigma_{m-1}, \quad \beta := \sigma_{m-1} \cdots \sigma_1, \quad \gamma := \sigma_1 \cdots \sigma_{m-2},$$

the full-twist identities are

$$\alpha^m = \beta^m = \Delta_m^2, \quad \gamma^{m-1} \beta \alpha = \Delta_m^2.$$

The first movie is

$$\begin{aligned} & \left(\begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} \right)^m \left(\begin{array}{c} \diagdown \diagup \\ \diagup \diagdown \end{array} \right)^m \alpha^{m=\beta^m} \left(\begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} \right)^m \left(\begin{array}{c} \diagdown \diagup \\ \diagup \diagdown \end{array} \right)^m \\ & \left(\begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} \right)^m \left(\begin{array}{c} \diagdown \diagup \\ \diagup \diagdown \end{array} \right)^m = \left(\begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} \right)^{m-1} \left(\begin{array}{c} \diagup \diagdown \diagup \diagdown \\ \diagdown \diagup \diagdown \diagup \end{array} \right) \left(\begin{array}{c} \diagdown \diagup \\ \diagup \diagdown \end{array} \right)^{m-1}. \end{aligned}$$

$$\begin{aligned}
& \left(\text{diagram} \right)^{m-1} \left(\text{diagram} \right)^{m-1} \left(\text{diagram} \right)^{m-1} \stackrel{\text{H}^3}{\sim}_H \left(\text{diagram} \right)^{m-1} \left(\text{diagram} \right)^{m-1} \left(\text{diagram} \right)^{m-1} \\
& \left(\text{diagram} \right)^{m-1} \left(\text{diagram} \right)^{m-1} \left(\text{diagram} \right)^{m-1} \stackrel{\text{H}^3}{\sim}_H \cdots \stackrel{\text{H}^3}{\sim}_H \left(\text{diagram} \right)^m \\
& \left(\text{diagram} \right)^m = \left(\left(\text{diagram} \right)^2 \right)^{m/2}.
\end{aligned}$$

Hence

$$\varphi_1^2 \sim_H \varphi_0^{m/2} = \varphi_0^{g+1}.$$

The cyclic move on the identity factorization T^2 , followed by the recursive full-twist relation, gives the second movie:

$$\begin{aligned}
& \left(\text{diagram} \right)^{2(m-1)} \left(\text{diagram} \right)^2 \stackrel{\text{cyc.}}{\sim}_H \left(\text{diagram} \right)^{m-1} \left(\text{diagram} \right)^{m-1} \left(\text{diagram} \right) \left(\text{diagram} \right) \\
& \left(\text{diagram} \right)^{m-1} \left(\text{diagram} \right)^{m-1} \left(\text{diagram} \right) \left(\text{diagram} \right) \stackrel{PQ=\alpha^m}{\sim}_H \left(\text{diagram} \right)^{m-1} \left(\text{diagram} \right)^m \left(\text{diagram} \right) \\
& \left(\text{diagram} \right)^{m-1} \left(\text{diagram} \right)^m \left(\text{diagram} \right) \stackrel{\Delta_m^2 \in Z(B_m)}{\sim}_H \left(\text{diagram} \right)^m \left(\text{diagram} \right)^{m-1} \left(\text{diagram} \right) \\
& \left(\text{diagram} \right)^m \left(\text{diagram} \right)^{m-1} \left(\text{diagram} \right) \stackrel{PQ=\alpha^m}{\sim}_H \left(\text{diagram} \right)^m \left(\text{diagram} \right)^m.
\end{aligned}$$

Therefore

$$\varphi_2 \varphi_0 \sim_H \varphi_1^2,$$

which completes all three equivalences. $\square$

2. TOPOLOGY OF LEFSCHETZ FIBRATIONS

Morse theory turns vanishing cycles into 2-handles, while monodromy substitutions turn relations in mapping class groups into surgeries on total spaces.

2.1. Handle Decompositions and Invariants. Throughout, $f: X \rightarrow \Sigma$ has genus g , vanishing cycles $c_1, \dots, c_n$, and critical values $\mathcal{C}$. Write

$$\Sigma_{g,r} := \text{the genus-}g \text{ surface with } r \text{ boundary components}, \quad \mathcal{M}_{g,r} := \pi_0(\text{Diff}^+(\Sigma_{g,r}, \partial\Sigma_{g,r})).$$

For the radial Morse model of [6], suppose $f: X \rightarrow D^2$ realizes $(t_{c_1}, \dots, t_{c_n})$ along

$$z_k := \frac{1}{2} \exp\left(\frac{2\pi\sqrt{-1}k}{n}\right), \quad R_k := \{tz_k : 0 \leq t \leq 1\}.$$

The complex-linear change $(z, w) \mapsto (z + \sqrt{-1}w, z - \sqrt{-1}w)$ identifies zw with $z^2 + w^2$. Fix $0 < \delta < \frac{1}{2}$, write $D_\delta := \{|z| \leq \delta\}$, and set

$$h(z) := |z|^2 - \frac{1}{4}, \quad \tilde{h} := h \circ f \quad \text{on } X - f^{-1}(\text{Int}(D_\delta)), \quad f^{-1}(D_\delta) \cong D_\delta \times \Sigma_g.$$

Proposition 2.1.1 (Index-two critical points, [6]). *The function*

$$\tilde{h}: X - f^{-1}(\text{Int}(D_\delta)) \longrightarrow \mathbb{R}$$

is Morse with critical points $p_1, \dots, p_n$, each of index 2, and

$$X \cong (D^2 \times \Sigma_g) \cup h_1^2 \cup \dots \cup h_n^2.$$

Proof. The regular locus contributes no critical points, while each complex-Morse point has real index two.

$$\begin{aligned} \tilde{d}h &= dh \circ df, \quad dh \neq 0, \quad df \text{ surjective off } \text{Crit}(f) \implies \text{Crit}(\tilde{h}) = \text{Crit}(f). \\ (\tilde{h} \circ \varphi_k^{-1})(z, w) &= -x_1^2 + y_1^2 - x_2^2 + y_2^2 \implies \text{ind}(p_k) = 2. \end{aligned}$$

The central product region and the outer regular collar contain no further critical points, so Morse theory gives the decomposition. $\square$

Proposition 2.1.2 (Vanishing-cycle attachment, [16, 6]). *The handle h_k^2 is attached along $\{2z_k\} \times c_k$ with framing -1 relative to the fiber framing.*

Proof. Compare the descending-disk framing with the normal framing inherited from the regular fiber.

$$\begin{aligned} S_k &= \{y_1 = y_2 = 0, x_1^2 + x_2^2 = \varepsilon\} \mapsto \{uv = \varepsilon, |u| = |v|\} = c_k, \\ v_1 &:= \frac{\partial}{\partial y_1}, \quad u := x_2 \frac{\partial}{\partial x_1} - x_1 \frac{\partial}{\partial x_2}, \quad v_2 := x_2 \frac{\partial}{\partial y_1} - x_1 \frac{\partial}{\partial y_2}, \\ \text{span}\{u, v_2\} &= Tf^{-1}(\varepsilon)|_{S_k}. \end{aligned}$$

Writing $x_1 = \sqrt{\varepsilon} \cos \theta$ and $x_2 = \sqrt{\varepsilon} \sin \theta$ gives

$$\begin{aligned} v_2(\theta) &= \sqrt{\varepsilon} \left(\sin \theta \frac{\partial}{\partial y_1} - \cos \theta \frac{\partial}{\partial y_2} \right), \quad \text{wind}(v_2, v_1) = 1, \\ \text{fr}_{\text{att}} - \text{fr}_F &= -1. \end{aligned}$$

Thus the attaching circle is c_k with relative framing -1 . $\square$

Proposition 2.1.3 (Kirby diagram rule, [16, 6]). *A Kirby diagram for X is obtained from that of $D^2 \times \Sigma_g$ by drawing the c_k in parallel levels ordered toward the reader by increasing k , each with relative framing -1 .*

Proof. The radial filtration places the Lefschetz handles in increasing angular order.

$$D^2 \times \Sigma_g \xrightarrow{(c_1, -1)} X_1 \xrightarrow{(c_2, -1)} \dots \xrightarrow{(c_n, -1)} X \quad \xrightarrow{2.1.1, 2.1.2} \quad c_1 < c_2 < \dots < c_n.$$

This is the stated Kirby diagram. $\square$

Example 2.1.4 (Genus-two and genus-one Kirby diagrams, [16, 6]). By Proposition 2.1.3, a chain $c_1, c_2, c_3 \subseteq \Sigma_2$ gives Figure ???. For $a, b \subseteq \Sigma_1$ with $i(a, b) = 1$, the words (t_a) and (t_a, t_b) give the fishtail and cusp neighborhoods after the standard slides and cancellations.

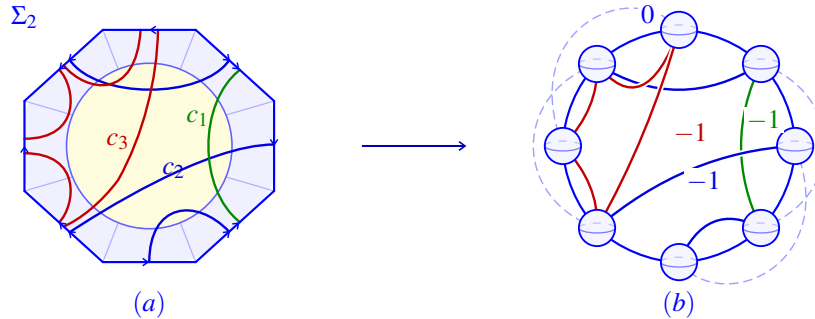


FIGURE 4. The curves $c_1, c_2, c_3 \subseteq \Sigma_2$ and the associated Kirby diagram; the exterior dashed arcs pair the eight spherical feet into four 1-handles.

Definition 2.1.5 (Section, [6]). A *section* of $f: X \rightarrow S^2$ is a smooth map

$$s: S^2 \longrightarrow X, \quad f \circ s = \text{id}_{S^2}.$$

Lemma 2.1.6 (Boundary twists produce sections, [6]). *If the identity factorization lifts to*

$$t_{c_1} \cdots t_{c_n} = t_{d_1}^{m_1} \cdots t_{d_r}^{m_r} \quad \text{in } \mathcal{M}_{g,r},$$

where the d_j are boundary-parallel and $m_j > 0$, then f admits disjoint sections $S_1, \dots, S_r$ with $S_j^2 = -m_j$.

Proof. The lift preserves r marked fiber disks, whose centers form disjoint sections, and the boundary twists record their normal clutching maps.

$$t_{c_1} \cdots t_{c_n} = \prod_{j=1}^r t_{d_j}^{m_j} \xrightarrow{1.1.12} s_j: S^2 \longrightarrow X,$$

$$\kappa_j: S^1 \longrightarrow \text{SO}(2), \quad \kappa_j(e^{\sqrt{-1}\theta}) = e^{-m_j \sqrt{-1}\theta}, \quad S_j^2 = e(\nu S_j) = \deg(\kappa_j) = -m_j.$$

Thus every marked disk supplies the asserted section and self-intersection. $\square$

Example 2.1.7 (Sections from the chain relations, [6]). The boundary-chain relations

$$(t_1 t_2 \cdots t_{2g+1})^{2g+2} = t_{d_1} t_{d_2} \quad \text{in } \mathcal{M}_{g,2}, \quad (t_1 t_2 \cdots t_{2g})^{4g+2} = t_d \quad \text{in } \mathcal{M}_{g,1},$$

give two disjoint (-1) -sections of f_1 and one (-1) -section of f_2 . By Figure ??, the boundary components are disjoint from the chain curves.

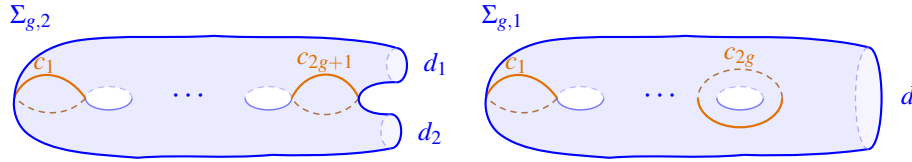


FIGURE 5. Standard curve configurations on the bordered surfaces $\Sigma_{g,2}$ and $\Sigma_{g,1}$.

Example 2.1.8 (Elliptic surfaces, [16, 6]). Let $a, b \subseteq \Sigma_1$ satisfy $i(a, b) = 1$. The fibration with factorization $(t_a, t_b)^{6n}$ is denoted $E(n)$, and

$$(t_a t_b)^{6n} = t_d^n \quad \text{in } \mathcal{M}_{1,1} \xrightarrow{2.1.6} \text{a section of square } -n.$$

Capping the disk fibration adds one 2-handle, two 3-handles, and one 4-handle; the final 2-handle is framed $-n$, and

$$E(1) \cong \mathbb{CP}^2 \# 9\overline{\mathbb{CP}}^2.$$

By Figure ??, the boundary twist supplies the displayed section handle.

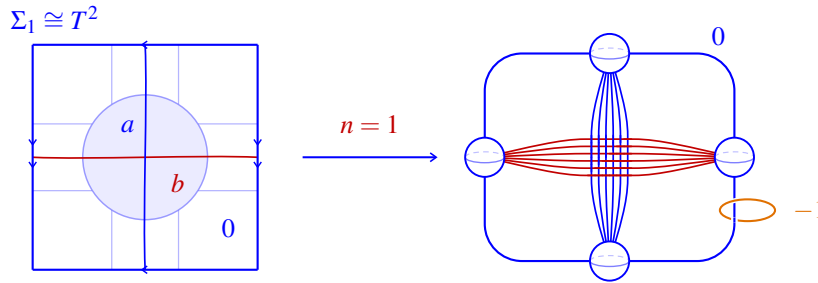


FIGURE 6. For $n = 1$, the curves $a, b \subseteq \Sigma_1$ produce six (-1) -framed handles of each type, while the linked meridian records the (-1) -section of $E(1)$.

Proposition 2.1.9 (Euler characteristic, [6]). *For $n = \#\mathcal{C}$,*

$$(3) \quad \chi(X) = \chi(\Sigma_g)\chi(\Sigma) + n.$$

Proof. Decompose the base into the regular complement and one disk about each critical value.

$$\begin{aligned} X^\circ &:= f^{-1}\left(\Sigma - \bigcup_i \text{Int}(D_i)\right), & X_i &:= f^{-1}(D_i), & X &= X^\circ \cup \bigcup_{i=1}^n X_i. \\ \chi(X^\circ) &= \chi(\Sigma_g)(\chi(\Sigma) - n), & \chi(X_i) &= \chi(\Sigma_g) + 1, & \chi(X^\circ \cap X_i) &= 0. \\ \chi(X) &= \chi(\Sigma_g)(\chi(\Sigma) - n) + n(\chi(\Sigma_g) + 1) = \chi(\Sigma_g)\chi(\Sigma) + n. \end{aligned}$$

This proves Equation (3). $\square$

Proposition 2.1.10 (Fundamental group, [6]). *If $\Sigma = S^2$, f admits a section, and*

$$N := \langle\langle c_1, \dots, c_n \rangle\rangle \trianglelefteq \pi_1(\Sigma_g),$$

then

$$\pi_1(X) \cong \pi_1(\Sigma_g)/N.$$

Proof. Compute over a disk containing the critical values and check that the complementary cap adds no relation.

$$\begin{aligned} S^2 &= D_+ \cup D_-, & W &:= f^{-1}(D_+) \cong (D^2 \times \Sigma_g) \cup h_1^2 \cup \dots \cup h_n^2, \\ W &\xrightarrow{\text{van Kampen}} \pi_1(W) \cong \pi_1(\Sigma_g)/N. \end{aligned}$$

The cap contributes only the following handles.

$$\begin{aligned} f^{-1}(D_-) &\text{ contributes } h^2 \cup (2g)h^3 \cup h^4, & \pi_1(X) &= \pi_1(W)/\langle\langle \ell \rangle\rangle, \\ \ell &= \partial(s(D_+)) = 1 \in \pi_1(W), & h^3, h^4 &\text{ preserve } \pi_1 \implies \pi_1(X) = \pi_1(W). \end{aligned}$$

Thus $\pi_1(X) \cong \pi_1(\Sigma_g)/N$. $\square$

Proposition 2.1.11 (Spin criterion, [21]). *Let the base be D^2 , or S^2 with a section S of square $-m$. The total space X is spin exactly when there is a quadratic refinement*

$$q: H_1(\Sigma_g; \mathbb{Z}/2) \longrightarrow \mathbb{Z}/2, \quad q(x+y) = q(x) + q(y) + x \cdot y, \quad q([c_i]) = 1$$

for every i , and, over S^2 , $m \equiv 0 \pmod{2}$.

Proof. Encode a fiber spin structure by its quadratic refinement and test extension over each remaining two-handle.

$$\begin{aligned} \text{Spin}(\Sigma_g) &\longleftrightarrow q, & \langle w_2(X), h_i^2 \rangle &= q([c_i]) + 1. \\ \text{extension across every } h_i^2 &\iff q([c_i]) = 1 \quad (1 \leq i \leq n). \\ \Sigma = S^2 &\implies \langle w_2(X), [S] \rangle \equiv S^2 \equiv m \pmod{2} \iff m \equiv 0 \pmod{2}. \end{aligned}$$

These are exactly the necessary and sufficient extension conditions. $\square$

Proposition 2.1.12 (Signature decomposition, [18, 20, 6]). *For a manifold with boundary, let σ denote the signature of its relative intersection form. With the disks D_i above,*

$$\begin{aligned} \sigma(X) &= \sigma\left(X - \bigcup_i \text{Int}(f^{-1}(D_i))\right) + \sum_{i=1}^n \sigma(f^{-1}(D_i)), \\ \sigma(f^{-1}(D_i)) &= \begin{cases} 0, & c_i \text{ nonseparating,} \\ -1, & c_i \text{ separating.} \end{cases} \end{aligned}$$

The regular term is computed from the Meyer cocycle on a pants decomposition of the base.

Proof. Cut along the singular-fiber mapping tori and compute the nondegenerate local forms.

$$\left(\begin{array}{ccc} & \bigsqcup_i \partial f^{-1}(D_i) & \\ \swarrow & & \searrow \\ X^\circ & & \bigsqcup_i f^{-1}(D_i) \end{array} \right) \xrightarrow{\text{Novikov additivity}} \sigma(X) = \sigma(X^\circ) + \sum_i \sigma(f^{-1}(D_i)).$$

$$c_i \text{ nonseparating} \implies \text{the local } H_2\text{-class is radical} \implies \sigma(f^{-1}(D_i)) = 0,$$

$$c_i = \partial F_h \implies A_i := F_h \cup \text{handle core}, \quad A_i^2 = -1,$$

$$Q_i^{\text{nd}} = [-1] \implies \sigma(f^{-1}(D_i)) = -1.$$

A pants decomposition evaluates $\sigma(X^\circ)$ by the Meyer cocycle, completing the decomposition. $\square$

Definition 2.1.13 (Hyperelliptic Lefschetz fibration, [5]). The fibration f is *hyperelliptic* when a hyperelliptic involution $\iota: \Sigma_g \rightarrow \Sigma_g$ satisfies

$$\rho_f(\pi_1(\Sigma - \mathcal{C}, q_0)) \subseteq C_{\mathcal{M}_g}([\iota]).$$

For $\Sigma = S^2$, an invariant-cycle factorization witnesses this condition. For positive-genus bases, the monodromies of the base-handle generators must also centralize $[\iota]$.

Theorem 2.1.14 (Hyperelliptic signature formula, [5]). Suppose $\partial \Sigma = \emptyset$. If $n_0(f)$ counts the nonseparating vanishing cycles and $n_h(f)$ counts the separating cycles cutting off genus h , then

$$\sigma(X) = -\frac{g+1}{2g+1}n_0(f) + \sum_{h=1}^{\lfloor g/2 \rfloor} \left(\frac{4h(g-h)}{2g+1} - 1 \right) n_h(f).$$

Proof. Endo's cobounding function localizes the Meyer cocycle on the hyperelliptic mapping class group [5].

$$\tau_g|_{C_{\mathcal{M}_g}([\iota])} = \delta \phi_g, \quad \sigma(X) = \sum_i \sigma_{\text{loc}}(c_i),$$

$$\sigma_{\text{loc}}(c_i) = \begin{cases} -\frac{g+1}{2g+1}, & c_i \text{ nonseparating,} \\ \frac{4h(g-h)}{2g+1} - 1, & c_i \text{ cuts off genus } h. \end{cases}$$

$$\sigma(X) = -\frac{g+1}{2g+1}n_0(f) + \sum_{h=1}^{\lfloor g/2 \rfloor} \left(\frac{4h(g-h)}{2g+1} - 1 \right) n_h(f).$$

Summing the local signatures gives the formula. $\square$

Example 2.1.15 (Invariants of the standard fibrations, [5, 6]). For the fibrations of Definition 1.2.9,

	n	σ	χ
f_0	$4(2g+1)$	$-4(g+1)$	$4(g+2)$
f_1	$2(g+1)(2g+1)$	$-2(g+1)^2$	$2(2g^2+g+3)$
f_2	$4g(2g+1)$	$-4g(g+1)$	$4(2g^2+1)$

by Equation (3) and Theorem 2.1.14. Moreover,

$$\langle\langle c_1, \dots, c_{2g+1} \rangle\rangle = \pi_1(\Sigma_g) \xrightarrow{2.1.10} \pi_1(X_i) = 1,$$

$$\text{boundary-twist parity or no compatible } q \xrightarrow{2.1.11} w_2(X_i) \neq 0 \quad (i = 0, 1, 2).$$

Definition 2.1.16 (Fiber sum, [13, 6]). Choose regular fibers

$$F = f^{-1}(q), \quad F' = (f')^{-1}(q'),$$

disks $D \ni q$, $D' \ni q'$, and product identifications

$$\Phi_0: \Sigma_g \times D \xrightarrow{\cong} f^{-1}(D), \quad \Phi'_0: \Sigma_g \times D' \xrightarrow{\cong} (f')^{-1}(D').$$

For $\Psi: \Sigma_g \rightarrow \Sigma_g$ orientation-preserving and $r: \partial D \rightarrow \partial D'$ orientation-reversing, remove the fiber neighborhoods and glue by

$$\Phi'_0 \circ (\Psi \times r) \circ \Phi_0^{-1}.$$

The resulting fibration is

$$f \#_{\Psi} f': X \#_{\Psi, F} X' \longrightarrow \Sigma \# \Sigma'.$$

Proposition 2.1.17 (Fiber-sum monodromy, [6]). *If f, f' lie over S^2 , have factorizations*

$$(t_{c_1}, \dots, t_{c_n}), \quad (t_{d_1}, \dots, t_{d_m}),$$

and $\Psi = [\Psi]$, then $f \#_{\Psi} f'$ has factorization

$$(t_{c_1}, \dots, t_{c_n}, \Psi^{-1} t_{d_1} \Psi, \dots, \Psi^{-1} t_{d_m} \Psi).$$

Proof. Deleting the gluing disks concatenates the Hurwitz systems, while the fiber identification conjugates the second one.

$$P_{d'_j}^{\text{new}} = \Psi^{-1} P_{d'_j}^{f'} \Psi, \quad \rho_{f \# f'}([a'_j]) = \Psi^{-1} t_{d_j} \Psi, \quad 1 \leq j \leq m.$$

Thus concatenation gives precisely the displayed factorization. $\square$

Example 2.1.18 (Stable exotic standard sums, [12, 13, 22, 24, 23]). For $g \geq 2$, $\#_F^g X_0$ and X_2 are simply connected and have the same Euler characteristic and signature. The fiber sum has a section of square $-g$; if g is even, the relation among the pairwise-disjoint odd chain curves gives

$$[c_1] + [c_3] + \dots + [c_{2g+1}] = 0, \quad q([c_i]) = 1 \implies 0 = g + 1 = 1 \quad \text{in } \mathbb{Z}/2.$$

For odd g , the section has odd square; for even g , the displayed relation obstructs q . By Proposition 2.1.11, both spaces are nonspin, and

$$\#_F^g X_0 \approx X_2 \approx Y := \#^{2g^2-2g+1} \mathbb{CP}^2 \# \#^{6g^2+2g+1} \overline{\mathbb{CP}}^2.$$

Freedman's theorem gives the homeomorphisms [12]. Both Lefschetz total spaces are symplectic with nonzero Seiberg–Witten invariants, whereas Y has vanishing invariant [13, 22, 24]. Moreover, Usher's theorem gives [23]

$$b_2^+ > 1, \quad \#_F^g X_0 \text{ smoothly minimal}, \quad X_2 \supseteq \text{a } (-1)\text{-section}, \\ \#_F^g X_0 \not\cong X_2.$$

The standard Hurwitz equivalence finally gives

$$\varphi_0^{g+1} \sim_H \varphi_2 \varphi_0 \xrightarrow{1.2.10, 2.1.17, 1.2.8} \#_F^{g+1} X_0 \cong X_2 \#_F X_0.$$

2.2. Substitutions and Charts. Throughout, a *relator* is a word in Dehn twists equal to 1, and

$$\chi_h := \frac{\sigma + \chi}{4}, \quad K^2 := 3\sigma + 2\chi, \quad \mathcal{M}_1 \cong \mathrm{SL}(2, \mathbb{Z}) = \langle t_1, t_2 \mid t_1 t_2 t_1 = t_2 t_1 t_2, (t_1 t_2)^6 = 1 \rangle.$$

Definition 2.2.1 (Monodromy substitution, [9]). Suppose an identity factorization contains a consecutive block $t_{a_1} \cdots t_{a_k}$, and

$$r := (t_{a_1} \cdots t_{a_k})^{-1} t_{b_1} \cdots t_{b_\ell} = 1.$$

The *substitution with respect to r* replaces that block by $t_{b_1} \cdots t_{b_\ell}$:

$$\phi = (t_1, \dots, t_p, t_{a_1}, \dots, t_{a_k}, t_{p+1}, \dots, t_{p+q}) \mapsto \phi' = (t_1, \dots, t_p, t_{b_1}, \dots, t_{b_\ell}, t_{p+1}, \dots, t_{p+q}).$$

The product of ϕ' is 1, so Theorem 1.2.2 realizes it by

$$f': X' \longrightarrow S^2.$$

Theorem 2.2.2 (Commutation and braid substitutions, [6]). *For*

$$r_c := (t_a t_b)^{-1} t_b t_a \quad (i(a, b) = 0), \quad r_b := (t_a t_b t_a)^{-1} t_b t_a t_b \quad (i(a, b) = 1),$$

the substituted fibration f' is isomorphic to f .



Proof. Both substitutions are single local Hurwitz equivalences.

$$i(a, b) = 0 \implies (t_a, t_b) \sim_H (t_a t_b t_a^{-1}, t_a) = (t_b, t_a),$$

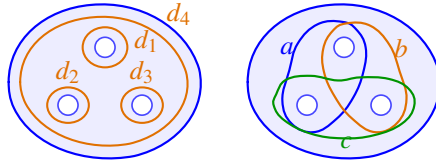
$$i(a, b) = 1 \implies (t_a, t_b, t_a) \sim_H (t_b, t_a, t_b).$$

$$\phi \sim_H \phi' \xrightarrow{1.2.8} f \cong f'.$$

Thus commutation and braid substitutions do not change the fibration. □

Lemma 2.2.3 (Lantern relation, [7]). *For the curves below in $\Sigma_{0,4}$,*

$$t_{d_1} t_{d_2} t_{d_3} t_{d_4} = t_a t_b t_c \quad \text{in } \mathcal{M}_{0,4}.$$



Theorem 2.2.4 (Lantern substitution as rational blowdown, [7]). *If a lantern substitution replaces $t_{d_1} t_{d_2} t_{d_3} t_{d_4}$ by $t_a t_b t_c$, then X' is the rational blowdown of X along a smoothly embedded (-4) -sphere.*

Proof. Compare the relative handlebodies Y_L, Y_R carried by the two sides of the lantern relation. By Figure ??, their handles transform as follows.

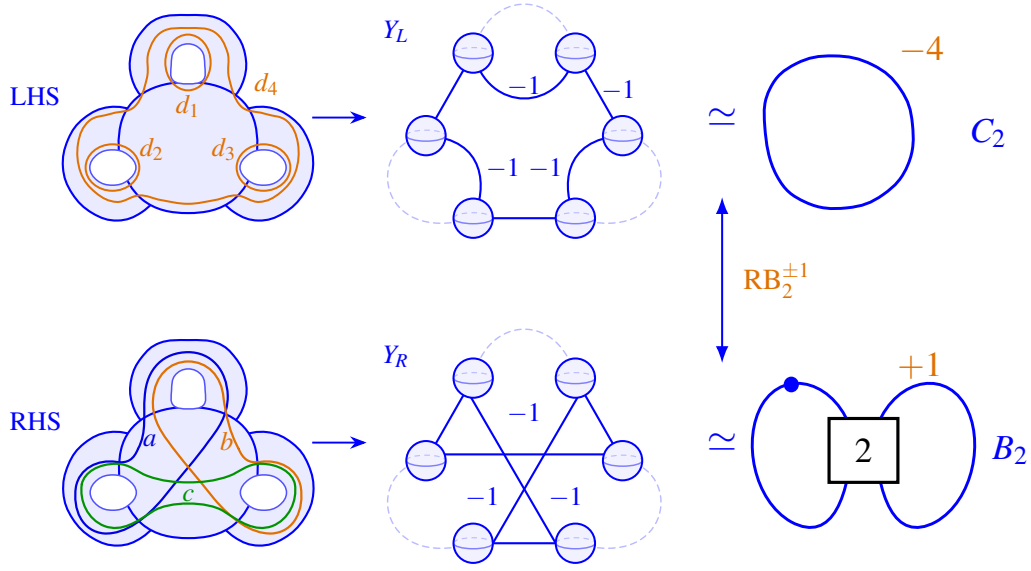


FIGURE 7. The lantern substitution and its Kirby presentations realize the rational-blowdown replacement of the (-4) -disk bundle C_2 by the rational ball B_2 .

The displayed slides and cancellations give

$$Y_L \cong C_2, \quad Y_R \cong B_2, \quad \partial B_2 = L(4, 1) = \partial C_2.$$

$$0 \longrightarrow C_2(B_2) = \mathbb{Z} \xrightarrow{\times 2} C_1(B_2) = \mathbb{Z} \longrightarrow 0,$$

$$H_2(B_2; \mathbb{Z}) = 0, \quad H_1(B_2; \mathbb{Z}) = \mathbb{Z}/2, \quad H_*(B_2; \mathbb{Q}) \cong H_*(D^4; \mathbb{Q}).$$

The substitution preserves the exterior.

$$X' \cong (X - \text{Int}(C_2)) \cup_{L(4,1)} B_2.$$

This is the rational blowdown along the zero-section of C_2 [11]. □

Proposition 2.2.5 (Signature change under a lantern substitution, [10]). *If X' is obtained from X by a lantern substitution, then*

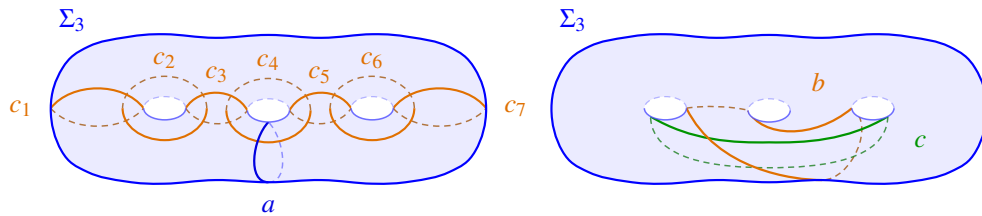
$$\sigma(X') = \sigma(X) + 1.$$

Proof. Use the rational-blowdown decomposition and Novikov additivity.

$$\sigma(X') = \sigma(X) - \sigma(C_2) + \sigma(B_2) = \sigma(X) - (-1) + 0 = \sigma(X) + 1.$$

Thus a lantern substitution raises the signature by one. □

Example 2.2.6 (A genus-three lantern substitution, [6]). Take the curves $c_1, \dots, c_7, a, b, c \subseteq \Sigma_3$ shown below.



Write $t_{c_i} = i$. Commutation and braid substitutions expose the lantern word:

$$\begin{aligned} \phi_1 &= (1234567)^8 \sim_H 12345 \cdot 12324 \cdot 56 \cdot 1357 \cdot 23 \cdot 46567 \cdot 34567 \cdot (1234567)^4, \\ &\xrightarrow{1357 \rightsquigarrow abc} 12345 \cdot 12324 \cdot 56 \cdot abc \cdot 23 \cdot 46567 \cdot 34567 \cdot (1234567)^4 =: \phi'_1. \end{aligned}$$

The resulting X'_1 has fifty-five vanishing cycles, and

$$\begin{aligned} \chi(X'_1) &= (-4) \cdot 2 + 55 = 47, & \sigma(X'_1) &= -32 + 1 = -31, \\ \langle\langle c_1, \dots, c_7 \rangle\rangle &= \pi_1(\Sigma_3) \xrightarrow{2.1.10} \pi_1(X'_1) = 1, \\ b_2(X'_1) &= \chi(X'_1) - 2 = 45, & b_2^+(X'_1) &= 7, & b_2^-(X'_1) &= 38. \\ \sigma(X'_1) &= -31 \not\equiv 0 \pmod{8} \implies Q_{X'_1} \text{ odd}, & \text{ks}(X'_1) &= 0. \end{aligned}$$

Freedman's theorem gives

$$X'_1 \approx \#_7 \mathbb{CP}^2 \#_{38} \overline{\mathbb{CP}}^2.$$

The left side has nonzero Seiberg–Witten invariant, while the connected sum has vanishing invariant, so the homeomorphism is not a diffeomorphism [12, 13, 22, 24, 7].

Example 2.2.7 (Baykur–Korkmaz–Simone fiber sum, [3, 23]). Write

$$i := t_{c_i}, \quad \bar{i} := t_{c_i}^{-1}, \quad \mathcal{I} := (7, 6, 5, 4, 3, 2, 1, 1, 2, 3, 4, 5, 6, 7, 7, 6, 5, 4, 3, 2, 1).$$

For

$$\phi_0 = (1, 2, 3, 4, 5, 6, 7, 7, 6, 5, 4, 3, 2, 1)^2,$$

Hurwitz moves expose a lantern block and produce

$$\begin{aligned} \phi_0 &\sim_H (1, 3, 5, 7, \bar{7}\bar{5}\bar{3}2357, \bar{7}\bar{5}457, \bar{7}67, \mathcal{I}), \\ &\xrightarrow{1357 \rightsquigarrow abc} (a, b, c, \bar{7}\bar{5}\bar{3}2357, \bar{7}\bar{5}457, \bar{7}67, \mathcal{I}) =: \phi'_0. \end{aligned}$$

For $Z := X_0 \#_F X'_0$,

$$\chi(Z) = 47, \quad \sigma(Z) = 2\sigma(X_0) + 1 = -31, \quad \pi_1(Z) = 1.$$

The sum is symplectically minimal [23]. A complex structure would make Z Kähler and complex minimal, while

$$K_Z^2 = 3\sigma(Z) + 2\chi(Z) = 1, \quad \chi_h(Z) = \frac{\sigma(Z) + \chi(Z)}{4} = 4, \quad q = 0, \quad p_g = 3.$$

$$K_Z^2 \geq 2p_g - 4 = 2, \quad K_Z^2 = 1 \implies \text{contradiction.}$$

Thus Z admits no complex structure by the Noether inequality [2]. Moreover,

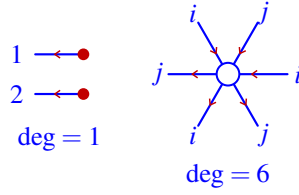
$$\frac{\chi(Z) - \sigma(Z)}{4} = \frac{47 + 31}{4} \notin \mathbb{Z}$$

excludes an almost-complex structure in the reverse orientation.

Definition 2.2.8 (Charts, [14]). A *chart* in an oriented surface Σ is a finite oriented graph

$$\Gamma \subseteq \text{Int}(\Sigma),$$

possibly with closed hoop components, whose edges are labeled 1 or 2. Its vertices have degree 1, 6, or 12, with the local models below. At a degree-six vertex three consecutive edges point inward and the other three outward, while a degree-one edge points outward in the chiral case.



Definition 2.2.9 (Intersection word, [14]). Let $\eta: [0, 1] \rightarrow \Sigma$ cross Γ transversely at $s_1, \dots, s_k$, with successive labels λ_i , $x_i := t_{\lambda_i}$, and

$$\varepsilon_i := \begin{cases} +1, & \text{the oriented edge crosses } \eta \text{ from left to right,} \\ -1, & \text{the oriented edge crosses } \eta \text{ from right to left.} \end{cases}$$

The *intersection word* is

$$w_\Gamma(\eta) := x_1^{\varepsilon_1} \cdots x_k^{\varepsilon_k}.$$

Proposition 2.2.10 (Chart monodromy, [14]). If C_Γ is the set of degree-one vertices and $q_0 \in \Sigma - \Gamma$, then

$$\rho_\Gamma: \pi_1(\Sigma - C_\Gamma, q_0) \longrightarrow \mathcal{M}_1, \quad \rho_\Gamma([\eta]) := [w_\Gamma(\eta)],$$

is a well-defined homomorphism.

Proof. A generic based homotopy crosses only an edge tangency or a degree-six or degree-twelve vertex.

$$\text{bigon} \longmapsto x_i x_i^{-1} = 1, \quad \text{deg 6} \longmapsto t_i t_j t_i = t_j t_i t_j,$$

$$\text{deg 12} \longmapsto (t_i t_j)^6 = 1.$$

$$w_\Gamma(\eta \eta') = w_\Gamma(\eta) w_\Gamma(\eta') \implies \rho_\Gamma([\eta \eta']) = \rho_\Gamma([\eta]) \rho_\Gamma([\eta']).$$

Hence the word depends only on the based homotopy class and defines a homomorphism. $\square$

Theorem 2.2.11 (Chart realization, [14]). Every genus-one Lefschetz fibration $f: X \rightarrow \Sigma$ with critical-value set C admits a chart satisfying

$$C_\Gamma = C, \quad \rho_\Gamma = \rho_f.$$

Conversely, every chiral chart whose degree-one meridians are conjugate to t_1 or t_2 determines a genus-one Lefschetz fibration.

Proof. Kamada's relation complex converts genus-one monodromy presentations into labeled oriented graphs and conversely [14].

$$\left(\begin{array}{ccc} & \text{dual relation complex} & \\ \text{Hurwitz presentation} & \xleftrightarrow{w_\Gamma} & \text{oriented chart } \Gamma \end{array} \right),$$

$$\text{deg 6} \longleftrightarrow t_1 t_2 t_1 = t_2 t_1 t_2, \quad \text{deg 12} \longleftrightarrow (t_1 t_2)^6 = 1,$$

$$\text{deg 1} \longleftrightarrow \text{a nodal meridian}.$$

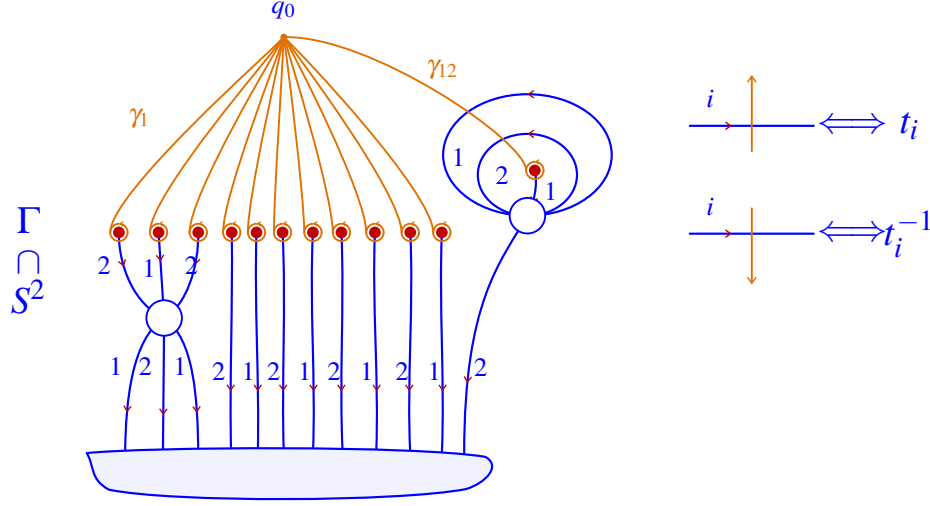
The edge label is 1 or 2, while its oriented crossing supplies the exponent in w_Γ .

$$\Gamma \longmapsto \text{torus bundle over } \Sigma - C_\Gamma \longmapsto \text{positive nodal fillings,}$$

$$C_\Gamma = C, \quad \rho_\Gamma = \rho_f.$$

These inverse constructions prove both assertions. $\square$

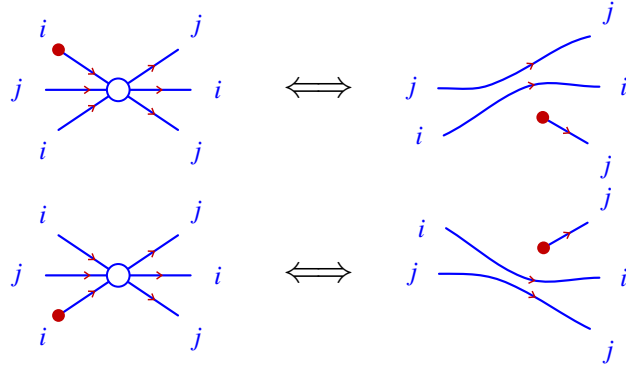
Example 2.2.12 (A chart on the sphere, [14]). The chart below has twelve degree-one vertices and one degree-twelve vertex.



Its meridian words are

$$\begin{aligned} & (t_2, t_1, t_2, t_2, t_1, t_2, t_1, t_2, t_1, t_2, t_1, t_1 t_2 t_1 t_2^{-1} t_1^{-1}), \quad t_1 t_2 t_1 t_2^{-1} t_1^{-1} = t_2, \quad t_2 t_1 t_2 = t_1 t_2 t_1 \\ & \implies \prod_{i=1}^{12} \rho_{\Gamma}([\gamma_i]) = (t_1 t_2)^6 = 1 \xrightarrow{2.2.11} f_{\Gamma}: X_{\Gamma} \longrightarrow S^2. \end{aligned}$$

Definition 2.2.13 (Chart moves, [14]). A *CI-move* is one of Kamada's standard word-preserving local replacements supported in a disk containing no degree-one vertex. A *CII-move* replaces either left-hand configuration below by the corresponding right-hand configuration, or conversely.



Theorem 2.2.14 (Invariance under chart moves, [14]). Suppose $\partial\Sigma \neq \emptyset$ or the represented genus-one fibrations have nonempty critical sets. If Γ_0 and Γ_1 are related by CI-moves, CII-moves, and isotopies, then

$$\rho_{\Gamma_0} \text{ is equivalent to } \rho_{\Gamma_1} \xrightarrow{1.2.4} f_0 \cong f_1.$$

Proof. Each local chart move preserves intersection words up to base and fiber identification.

$$\Gamma_0 \xrightarrow{\text{CI, CII, isotopy}} \Gamma_1 \implies w_{\Gamma_1}(\theta\eta) = \phi^{-1} w_{\Gamma_0}(\eta)\phi,$$

$$\rho_{\Gamma_0} \text{ equivalent to } \rho_{\Gamma_1} \xrightarrow{1.2.4} f_0 \cong f_1.$$

Thus chart moves preserve the represented fibration. □

Example 2.2.15 (Hurwitz equivalence seen by chart moves, [14]). Let

$$\phi_0 := (t_1, t_2)^6, \quad \phi'_0 := (t_1, t_1^{-1} t_2 t_1)^6 = \phi_0^{t_1}.$$

Then

$$\phi_0 \cdot \phi'_0 = (t_1, t_2)^6 \cdot (t_1, t_1^{-1} t_2 t_1)^6 \sim_H (t_1, t_2)^6 \cdot (t_2, t_1)^6 \sim_H \phi_0 \cdot \phi_0.$$

The CII configurations above realize the same equivalence directly on the charts of the two fiber sums.

Theorem 2.2.16 (Moishezon–Kas classification, [19, 15, 14]). *Two relatively minimal genus-one Lefschetz fibrations over S^2 with nonempty critical sets satisfy*

$$f \cong f' \iff \#\mathcal{C} = \#\mathcal{C}', \quad \#\mathcal{C}, \#\mathcal{C}' \in 12\mathbb{Z}_{>0}.$$

Proof. Abelianization determines the number of singular fibers, and the Moishezon–Kas normal form determines the factorization.

$$H_1(\mathcal{M}_1; \mathbb{Z}) \cong \mathbb{Z}/12, \quad [t_c] = 1, \quad \prod_{i=1}^{\#\mathcal{C}} t_{c_i} = 1, \\ \#\mathcal{C} = 12n.$$

The positive normal form gives

$$\phi \sim_H (t_1, t_2)^{6n}, \quad \phi' \sim_H (t_1, t_2)^{6n'}, \quad \#\mathcal{C} = \#\mathcal{C}' \iff n = n'. \\ n = n' \implies \phi \sim_H \phi' \xrightarrow{1.2.8} f \cong f'.$$

Hence the singular-fiber count is the complete isomorphism invariant. $\square$

Theorem 2.2.17 (Stable classification, [1, 8]). *For every $g > 2$ there is a universal chiral genus- g Lefschetz fibration*

$$f_0: M_0 \longrightarrow S^2.$$

For genus- g fibrations $f: X \rightarrow \Sigma$ and $f': X' \rightarrow \Sigma$ over a closed oriented surface, let n_0 count nonseparating cycles and n_h the separating cycles cutting off genus h . Then

$$\exists N \geq 0: f \# N f_0 \cong f' \# N f_0$$

if and only if

$$n_0(f) = n_0(f'), \quad n_h(f) = n_h(f') \quad \left(1 \leq h \leq \left\lfloor \frac{g}{2} \right\rfloor\right), \quad \sigma(X) = \sigma(X').$$

Proof. Package the stable additive data as

$$I(f) := (n_0(f), n_1(f), \dots, n_{\lfloor g/2 \rfloor}(f), \sigma(X)), \\ I(f \# N f_0) = I(f) + NI(f_0).$$

$$f \# N f_0 \cong f' \# N f_0 \implies I(f) + NI(f_0) = I(f') + NI(f_0) \implies I(f) = I(f').$$

By the universal-chart normal form of [8, Theorem 4.2],

$$I(f) = I(f') \implies \exists N \geq 0: f \# N f_0 \cong f' \# N f_0.$$

This proves the equivalence, extending Auroux’s sphere-base form [1]. $\square$

Part 2. Exercises

3. EXERCISES

Exercise 3.0.1 (1.1.3). Show that $f: X \rightarrow \Sigma$ can be deformed through smooth maps preserving the Lefschetz local models so that

$$f|_{\text{Crit}(f)}: \text{Crit}(f) \longrightarrow \Sigma$$

is injective.

Exercise 3.0.2 (1.1.6). (1) Show that $\rho_f([a])$ is independent of the representative a and trivialization Ψ .

(2) For the monodromy ρ'_f determined by another orientation-preserving $\Phi': \Sigma_g \rightarrow f^{-1}(q_0)$, show

$$\phi := [(\Phi')^{-1} \circ \Phi] \implies \rho'_f([a]) = \phi^{-1} \cdot \rho_f([a]) \cdot \phi.$$

(3) Show

$$\rho_f(\alpha\beta) = \rho_f(\alpha)\rho_f(\beta) \quad (\alpha, \beta \in \pi_1(\Sigma - \mathcal{C}, q_0)).$$

Exercise 3.0.3 (1.1.10). Let Ψ be the trivialization of Proposition 1.1.9, and set

$$f^{-1}(q_0) = \{(z, z^{-1}) \in \mathbb{C}^2 \mid z \neq 0\}, \quad \Psi_1(x) := \text{pr}_2(\Psi(1, x)).$$

(1) Show

$$\rho_f([a]) = [\Psi_1].$$

(2) For $z \in \mathbb{C} - \{0\}$, show

$$(\text{pr}_1 \circ \Psi_1 \circ \text{pr}_1^{-1})(z) = \begin{cases} z, & |z| \geq 2 \text{ or } |z| \leq \frac{1}{2}, \\ ze^{-\rho(|z|)\sqrt{-1}}, & \frac{1}{2} \leq |z| \leq 2, \end{cases}$$

where ρ is defined in Proposition 1.1.9. Identify this map as the right-handed Dehn twist along $\{|z| = 1\}$, corresponding to $Q^{-1}(1) \cap T$ at $(0, 0)$.

Exercise 3.0.4 (1.2.1). (1) Show that the one-vanishing-cycle construction in Theorem 1.2.1 is a genus- g Lefschetz fibration with monodromy t_c relative to γ and a suitable Φ .

(2) Extend the construction to $n > 1$.

Exercise 3.0.5 (1.2.2). Prove Theorem 1.2.2.

Exercise 3.0.6 (1.2.8). For $g \geq 2$ or $n \geq 1$, show

$$\left\{ \begin{array}{c} \text{genus-}g \text{ Lefschetz fibrations over } S^2 \\ \text{with } n \text{ singular fibers} \end{array} \right\} / \cong \longleftrightarrow \left\{ \begin{array}{c} t_{c_i} \text{ right-handed on } \Sigma_g \ (1 \leq i \leq n), \\ (t_{c_1}, \dots, t_{c_n}) \text{ modulo Hurwitz equivalence} \\ t_{c_1} \cdots t_{c_n} = 1 \end{array} \right\}.$$

Exercise 3.0.7 (1.2.10). (1) For $g = 1$, show

$$\varphi_0 \sim_H \varphi_1 \sim_H \varphi_2.$$

(2) Show

$$\varphi_0^{g+1} \sim_H \varphi_1^2 \sim_H \varphi_2 \varphi_0.$$

A proof for $g = 2$ suffices.

Exercise 3.0.8 (2.1.4). For $a, b \subseteq \Sigma_1$ with $i(a, b) = 1$, draw and simplify the Kirby diagrams of the genus-one disk fibrations determined by

$$(t_a), \quad (t_a, t_b).$$

Exercise 3.0.9 (2.1.9). Show

$$\chi(X) = \chi(\Sigma_g)\chi(\Sigma) + n.$$

Exercise 3.0.10 (2.2.2). For

$$r \in \{(t_a t_b)^{-1} t_b t_a, (t_a t_b t_a)^{-1} t_b t_a t_b\},$$

show

$$f \cong f'.$$

Exercise 3.0.11 (2.2.3). Verify

$$t_{d_1} t_{d_2} t_{d_3} t_{d_4} = t_a t_b t_c \quad \text{in } \mathcal{M}_{0,4}.$$

Exercise 3.0.12 (2.2.4). Using Figure ??, verify

$$Y_L \cong C_2, \quad Y_R \cong B_2.$$

Exercise 3.0.13 (2.2.10). Show that the map ρ_Γ of Example 2.2.12 is well-defined.

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